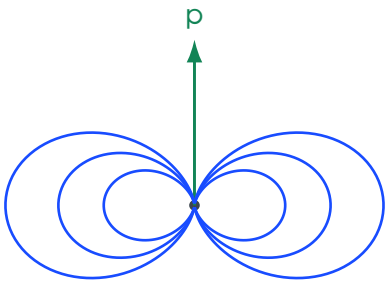
偶极场线：微分方程与极冠通量给出同一曲线族



$$r = C\sin^2\theta$$

$$\frac{dr}{r\,d\theta} = \frac{E_r}{E_\theta} = 2\cot\theta$$

$$\Psi(r,\theta) = \int_0^\theta E_r r^2 \sin\theta'\,d\theta' = \frac{kp}{r}\sin^2\theta$$

沿场线的极冠通量保持不变；
两种推导得到同一曲线常数。